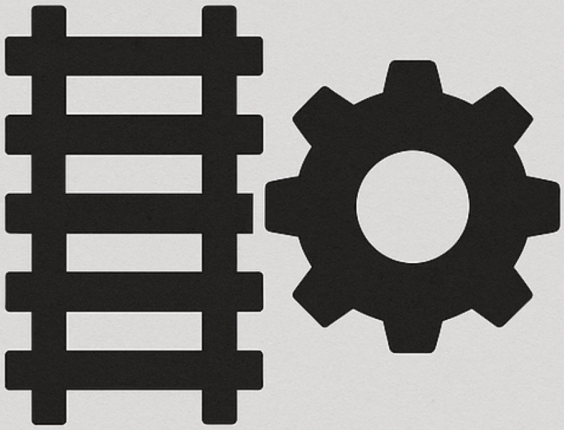
Texture Workshop User Guide



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Pete Willard

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Chapter 1. Introduction

Texture Workshop converts Microsoft Train Simulator (MSTS) and Open Rails textures between the file types you need when editing them.

Conversion	Use it when
ACE to editable PNG	You want to edit an existing route or rolling-stock texture in a normal image editor.
ACE directly to DDS	You want Open Rails DDS textures and do not need to edit anything first.
PNG/TGA/BMP to DDS	You have edited or new images and want DDS textures for Open Rails.
PNG/TGA/BMP to ACE	You have edited or new images and want ACE textures, which both MSTS and Open Rails can use.



MSTS can only use ACE textures. Open Rails can use ACE or DDS. If a texture must work in both, make it an ACE file.

Texture Workshop was previously called **DDS Workshop**. The name changed when PNG to ACE conversion was added. Your DDS Workshop settings are picked up automatically the first time you start Texture Workshop.

This guide describes Texture Workshop version 0.4.1.

Chapter 2. What Is Included

The release ZIP contains one folder. Keep everything in it together.

File or folder	What it is
`TextureWorkshop.exe`	The Texture Workshop program. This is the file you run.
`ace2png.exe`	Helper that turns ACE files into PNG files.
`png2dds.exe`	Helper that turns PNG files into DDS files.
`png2ace.exe`	Helper that turns PNG, TGA and BMP files into ACE files.
`ImageMagick\`	A portable copy of ImageMagick, used when making DDS files.
`Documentation\`	This user guide.
`README.md`	Short quick-start notes.
`THIRD_PARTY_LICENSES\`	License notices for ImageMagick.
`SHA256SUMS.txt`	Checksums you can use to confirm the download is complete.

The helper programs can also be used on their own from a Command Prompt. See Command-Line Tools.

Chapter 3. Installing

1. Download the Texture Workshop release ZIP.
2. Extract the whole ZIP to a normal folder, for example `C:\Tools\TextureWorkshop`.

   Do not run the program from inside the ZIP file.
3. Open the folder and double-click `TextureWorkshop.exe`.

You do not need to install Python. You do not need to install ImageMagick, because the release includes it in the `ImageMagick` folder.



Keep `TextureWorkshop.exe`, the three helper `.exe` files and the `ImageMagick` folder together in the same folder. If one is missing, Texture Workshop tells you which file it cannot find when you click **Run Conversion**.

3.1. If you install ImageMagick yourself

Only the DDS conversions use ImageMagick. If your copy of Texture Workshop does not have the `ImageMagick` folder, install ImageMagick from https://imagemagick.org/script/download.php#windows and turn on **Add application directory to your system path** during installation.

Texture Workshop looks for ImageMagick in this order:

1. The ImageMagick path set in **File** > **Settings**
2. `ImageMagick\magick.exe` beside `TextureWorkshop.exe`
3. `magick.exe` beside `TextureWorkshop.exe`
4. `magick.exe` on the system PATH

Chapter 4. The Main Window

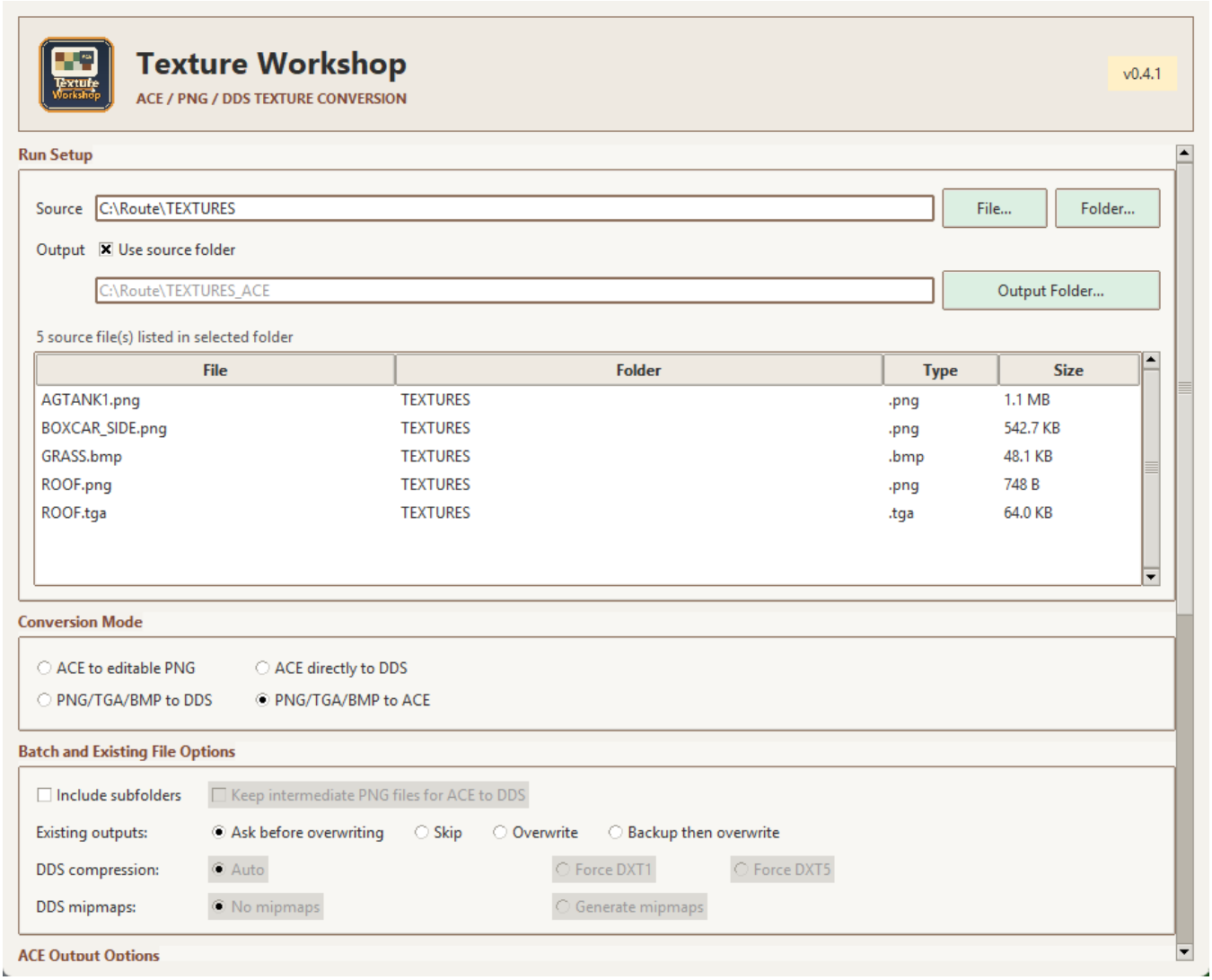


Figure 1. Texture Workshop main window

The window is divided into panels from top to bottom. If the window is small, scroll down to see the lower panels.

Run Setup

Choose the **Source**: one texture file (**File…**) or a whole folder (**Folder…**). The list underneath shows the files that will be converted in the current mode. Choose where the new files go: tick **Use source folder** to put them beside the originals, or untick it and pick an **Output Folder…**.

Conversion Mode

Choose one of the four conversions. The file list updates to show only the files that mode can

use.

Batch and Existing File Options

Include subfolders converts files in folders inside the source folder too. **Existing outputs** controls what happens when a file you are about to create already exists. See Existing Files and Backups. The DDS settings are only available in the two DDS modes.

ACE Output Options

The ACE format and ACE mipmap settings. These are only available in **PNG/TGA/BMP to ACE** mode.

Status

Run Conversion starts the batch. **Abort** stops it after the file being converted now is finished.

Output Log

Shows each file as it is converted, and a summary at the end. See Reading the Output Log.

Chapter 5. Converting Textures

Every conversion uses the same steps:

1. Choose the **Source** file or folder.
2. Choose the **Conversion Mode**.
3. Tick **Include subfolders** if you want files in subfolders converted too.
4. Choose the output location: **Use source folder**, or an **Output Folder**.
5. Check the options for the mode you picked (described below).
6. Click **Run Conversion**.
7. Answer any questions Texture Workshop asks about existing files or same-name files.
8. Read the summary at the end of the **Output Log**.

When you use a separate output folder together with **Include subfolders**, the subfolder layout is copied into the output folder. This keeps files with the same name in different folders apart.



Try a new setting on a copy of one or two textures first, then run the whole folder.

5.1. ACE to editable PNG

Makes a PNG file from each ACE file. Open the PNG in your image editor, make your changes, and save it as PNG again. Then convert it back with **PNG/TGA/BMP to DDS** or **PNG/TGA/BMP to ACE**.

This mode does not use ImageMagick.

5.2. ACE directly to DDS

Makes a DDS file from each ACE file in one step. Texture Workshop creates a temporary PNG first and deletes it afterwards.

Tick **Keep intermediate PNG files for ACE to DDS** if you also want to keep those PNG files. They are saved beside the DDS files.

The DDS compression and DDS mipmap settings apply. See the next section.

5.3. PNG/TGA/BMP to DDS

Makes a DDS file from each PNG, TGA or BMP file. TGA and BMP files are converted to a temporary PNG first.

DDS compression

- **Auto** (recommended): DXT1 for images without transparency, DXT5 for images with transparency.
- **Force DXT1**: always DXT1. Any transparency is lost.
- **Force DXT5**: always DXT5.

DDS mipmaps

- **No mipmaps**: the default.
- **Generate mipmaps**: adds smaller copies of the texture inside the DDS file, which can make distant textures look smoother.

5.4. PNG/TGA/BMP to ACE

Makes an ACE file from each PNG, TGA or BMP file. ACE files work in both MSTS and Open Rails. This mode does not use ImageMagick.

ACE format	What it does
Structured (safest)	Uncompressed. Works with the widest range of programs. Files are larger. This is the default.
DXT1	Compressed, for textures without transparency.
DXT3	Compressed, for textures with transparency.
DXT5	Compressed, for textures with transparency. Usually better than DXT3 for soft edges.
Auto DXT	Picks DXT1 for images without transparency and DXT5 for images with transparency.

Transparency (alpha)

Texture Workshop only writes a transparency channel when the image really uses transparency. Tiny amounts of almost-opaque transparency are treated as fully opaque, because some older MSTS ACE readers show stripes when an unneeded transparency

channel is present.

DXT3 and DXT5 need transparency

If you choose DXT3 or DXT5 for an image that has no transparency channel, that image is skipped and no ACE file is written for it. The log shows the reason. Any existing ACE file with that name is left untouched. Use **Auto DXT** if your folder has a mix of images with and without transparency.

ACE mipmaps

Generate mipmaps only works for square images whose sides are a power of two, such as 256 x 256, 512 x 512 or 1024 x 1024. Other images fail with an error in the log and no ACE file is written for them.

Chapter 6. Existing Files and Backups

Texture Workshop never replaces an existing file unless you have chosen to allow it. The **Existing outputs** setting decides what happens:

Setting	What happens to a file that already exists
Ask before overwriting	Texture Workshop asks you once, before converting anything. This is the default.
Skip	The existing file is kept and that source file is not converted.
Overwrite	The existing file is replaced. No backup is kept.
Backup then overwrite	A backup copy of the existing file is made, then it is replaced.

With **Ask before overwriting**, Texture Workshop checks all the files in the batch before it starts. If any of the new files already exist, it lists them:

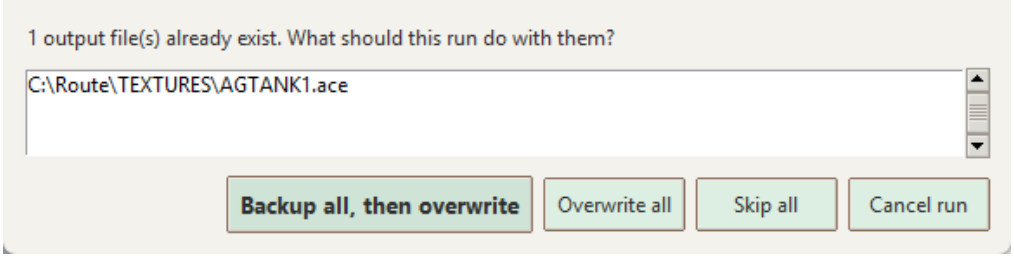


Figure 2. Existing output files question

- **Backup all, then overwrite**: back up every listed file, then replace it.
- **Overwrite all**: replace every listed file without a backup.
- **Skip all**: keep every listed file and do not convert the sources that would replace them.
- **Cancel run**: stop. Nothing is converted.

If none of the new files exist yet, no question is shown.

Backups are kept in the same folder as the file, with the date and time and `.bak` added to the name. For example:

```
AGTANK1.ace
AGTANK1.ace.20261024-075846.bak
```

To restore a backup, delete or rename the newer file, then remove the date and `.bak` from the backup's name.

Chapter 7. Same-Name Source Files

Two different source files can make an output file with the same name. For example, `ROOF.png` and `ROOF.tga` would both make `ROOF.ace`. Texture Workshop finds these before it starts and asks which one to use:

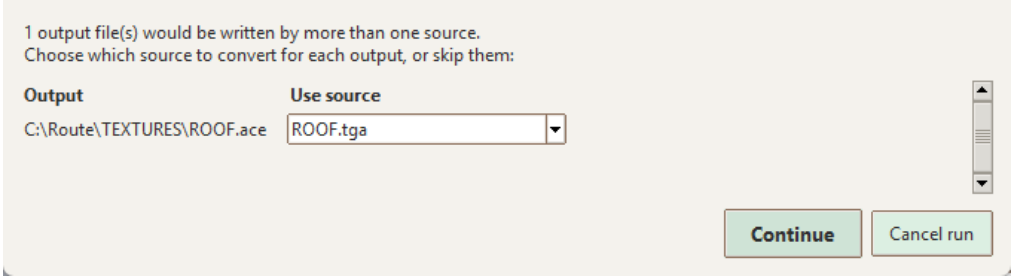


Figure 3. Same-name source files question

For each output file, choose the source file to convert, or **Skip all of these** to convert neither. **Skip all of these** is selected to begin with, so nothing is chosen for you. Click **Continue** to go on, or **Cancel run** to stop without converting anything.

Upper and lower case do not matter: `Roof.png` and `ROOF.tga` also count as the same name.

Chapter 8. Reading the Output Log

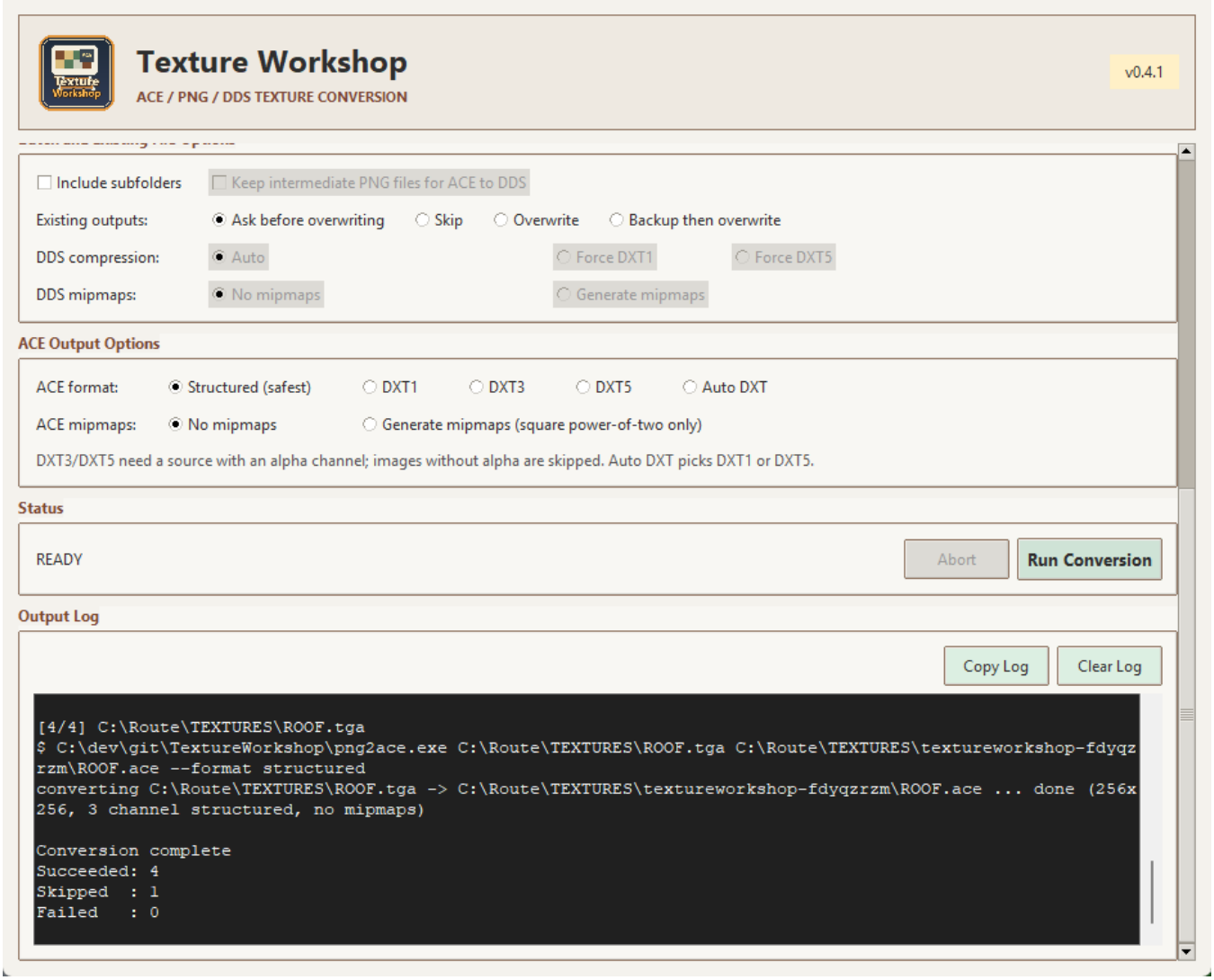


Figure 4. Output log after a run

The log lists each file as it is converted, including the messages from the helper programs. It also records the choices you made before the run, such as which same-name file was used.

At the end there is a summary:

Succeeded

New files that were written.

Skipped

Files that were not converted on purpose. For example, the output already existed and you chose **Skip**, you did not pick a same-name source file, or DXT3/DXT5 was chosen for an image without transparency.

Failed

Files that could not be converted. Scroll up in the log to find the error for each one.

Use **Copy Log** to copy the whole log to the clipboard, for example to paste it into a forum post when asking for help. **Clear Log** empties the log.

Chapter 9. Settings

Open **File** > **Settings...** to change:

- the location of `ace2png.exe`, `png2dds.exe` and `png2ace.exe`
- the location of ImageMagick `magick.exe`
- the **Theme** (Light or Dark)

Normally you do not need to change the program locations, because Texture Workshop finds the copies in its own folder.

Texture Workshop remembers your source, output, mode and option choices between sessions. They are stored in `%APPDATA%\TextureWorkshop\settings.ini`. If that file does not exist yet but DDS Workshop settings do, the DDS Workshop settings are used.

Chapter 10. Editing Tips

- Any image editor that saves PNG files will work, for example GIMP, Paint.NET, Affinity Photo or Photoshop.
- If the original texture had transparent areas, make sure your editor keeps the transparency when you save. A checkerboard background in the editor usually means the image has transparent areas.
- Keep texture sizes to powers of two, such as 256, 512, 1024 or 2048 pixels on each side. This is what MSTS and Open Rails expect, and it is required for ACE mipmaps.
- Work on copies of route and rolling-stock textures, not on the only copy you have.

Chapter 11. Troubleshooting

Problem	Likely cause	What to do
A message says required support files are missing	A helper program or the `ImageMagick` folder is not beside `TextureWorkshop.exe`	Extract the whole release ZIP again, or set the paths in **File** > **Settings**.
The file list is empty	The source folder has no files that the current mode can use, or the files are in subfolders	Check the **Conversion Mode**. Tick **Include subfolders** if the textures are in subfolders.
An image was skipped when making ACE files	DXT3 or DXT5 was chosen, and the image has no transparency	Choose **Auto DXT**, **DXT1** or **Structured** for that image.
An ACE file failed with a mipmap error	**Generate mipmaps** was chosen for an image that is not square and a power of two	Resize the image, or choose **No mipmaps**.
Transparency is missing	The image was saved without transparency, or **Force DXT1** was used for DDS	Save the image again with transparency kept. For DDS, use **Auto** or **Force DXT5**.
Nothing happened to some files	The output files already existed and **Skip** was chosen	Check the log for `SKIP` lines. Run again and choose **Backup all, then overwrite**.
Files went to the wrong folder	**Use source folder** or the output folder was not set as intended	Check the **Output** settings in **Run Setup** before you click **Run Conversion**.

Chapter 12. Command-Line Tools

The three helper programs can be used on their own from a Command Prompt. This is optional. Most users only need Texture Workshop.



If a folder name contains spaces, put quotes around the path. For example: `"C:\Open Rails\My Route\Textures"`

The examples assume the tools are in `C:\Tools\TextureWorkshop`.

12.1. ace2png.exe: ACE to PNG

Convert one ACE file. The PNG is written beside it:

```
C:\Tools\TextureWorkshop\ace2png.exe texture.ace
```

Convert all ACE files in a folder into another folder:

```
C:\Tools\TextureWorkshop\ace2png.exe ace_folder -o png_folder
```

If a PNG with the same name exists, it is renamed with `.bak` added before the new file is written.

12.2. png2dds.exe: PNG to DDS

png2dds.exe needs ImageMagick on the system PATH when it is used on its own.

Convert one PNG file. With no destination, the DDS file is written into a `dds` folder:

```
C:\Tools\TextureWorkshop\png2dds.exe texture.png
```

Convert all PNG files in a folder, optionally including subfolders:

```
C:\Tools\TextureWorkshop\png2dds.exe png_folder --destination dds_folder
C:\Tools\TextureWorkshop\png2dds.exe png_folder --destination dds_folder --recursive
```

Choose the compression or generate mipmaps:

```
C:\Tools\TextureWorkshop\png2dds.exe texture.png --destination texture.dds
```

```
--compression dxt1
C:\Tools\TextureWorkshop\png2dds.exe texture.png --destination texture.dds
--compression dxt5
C:\Tools\TextureWorkshop\png2dds.exe texture.png --destination texture.dds --mipmaps
generate
```

`--compression` can be `auto` (the default), `dxt1` or `dxt5`. If a DDS file with the same name exists, it is renamed with `.bak` added before the new file is written.

12.3. png2ace.exe: PNG, TGA or BMP to ACE

Convert one image. The ACE file is written beside it:

```
C:\Tools\TextureWorkshop\png2ace.exe texture.png
```

Convert one image to a chosen ACE file name:

```
C:\Tools\TextureWorkshop\png2ace.exe texture.png texture.ace
```

Convert all PNG and TGA files in a folder (not subfolders) into another folder:

```
C:\Tools\TextureWorkshop\png2ace.exe png_folder ace_folder
```

Choose the ACE format or generate mipmaps:

```
C:\Tools\TextureWorkshop\png2ace.exe texture.png texture.ace --format dxt1
C:\Tools\TextureWorkshop\png2ace.exe texture.png texture.ace --format auto-dxt
C:\Tools\TextureWorkshop\png2ace.exe texture.png texture.ace --mipmaps
```

`--format` can be `structured` (the default), `dxt1`, `dxt3`, `dxt5` or `auto-dxt`.



Used on its own, png2ace.exe replaces an existing ACE file without making a backup. Use Texture Workshop if you want to be asked first.

Chapter 13. Credits

ImageMagick is a separate third-party program and remains subject to its own license terms. See the `THIRD_PARTY_LICENSES` folder.

Referenced Open Rails ACE format documentation © Open Rails Project contributors.

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